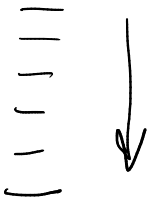
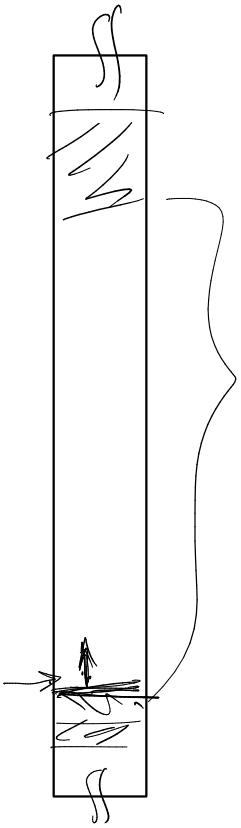
malloc(0);

```
p = malloc(0);
if(p == NULL) {
    exit(0);
}
```



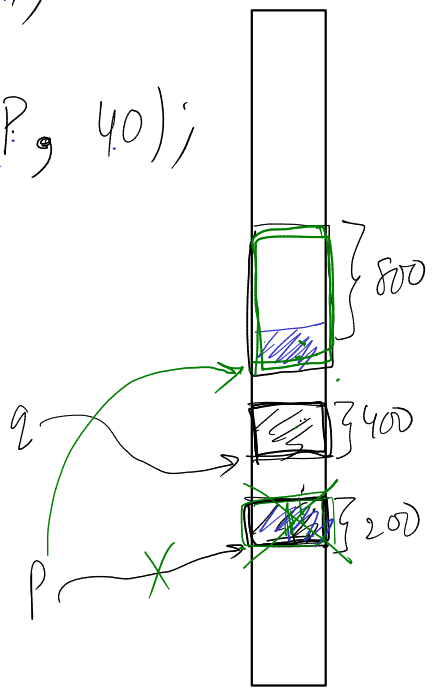


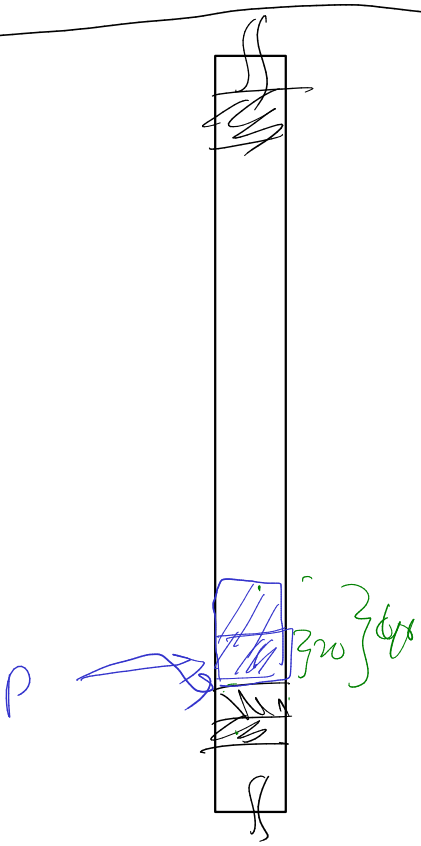
p = malloc(0);

↳ sbrk + mmap

```
p = malloc(20);
p = realloc(p, 40);
```

```
p = malloc(-);
q = malloc( );
p = realloc(p, 800);
```





2D - Array

```
int arr[][]; // Error.
```

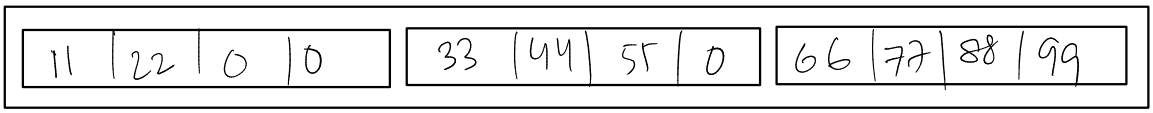
```
int arr[][] = {{11,22,33},{44,55,66}}; // Error
```

```
int arr[][4] = {{11,22},{33,44,55},{66,77,88,99}};
```

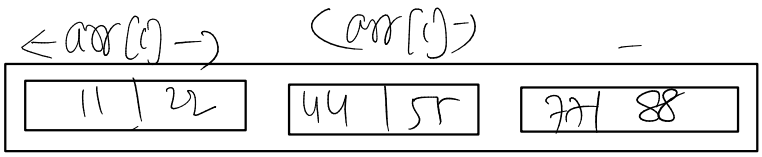
sizeof(arr) => 48

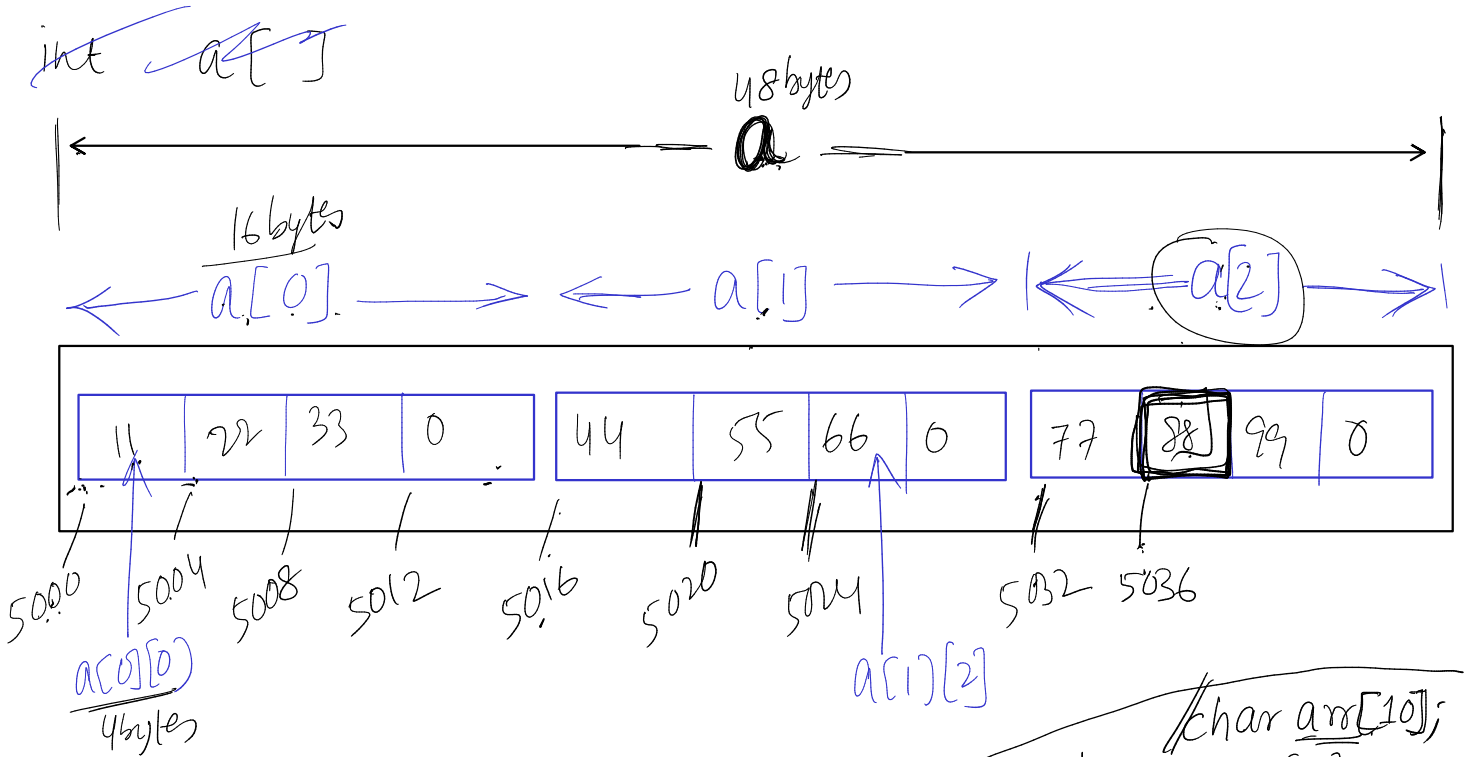
sizeof(*arr) => 16

sizeof(**arr) => 4




```
int arr[3][2] = {{11,22,33},{44,55,66},{77,88},{99,100}};
```





%lu", a) // 5000

%lu %lu %lu", a+1, a[0]+1, a[0][0]+1);
5016 5004 12

%lu %lu %lu", *a+1, **a+1, *a[1]+1
a[0]+1 a[0][0]+1, a[1][0]+1
5004 12 45

%lu", **a+1+1 => a[0][0]+2

%lu, *(*(a+1)+1) => a[1][1]

%lu, &a[2][1]); // 5036
a[2]+1 → *(a+2)+1

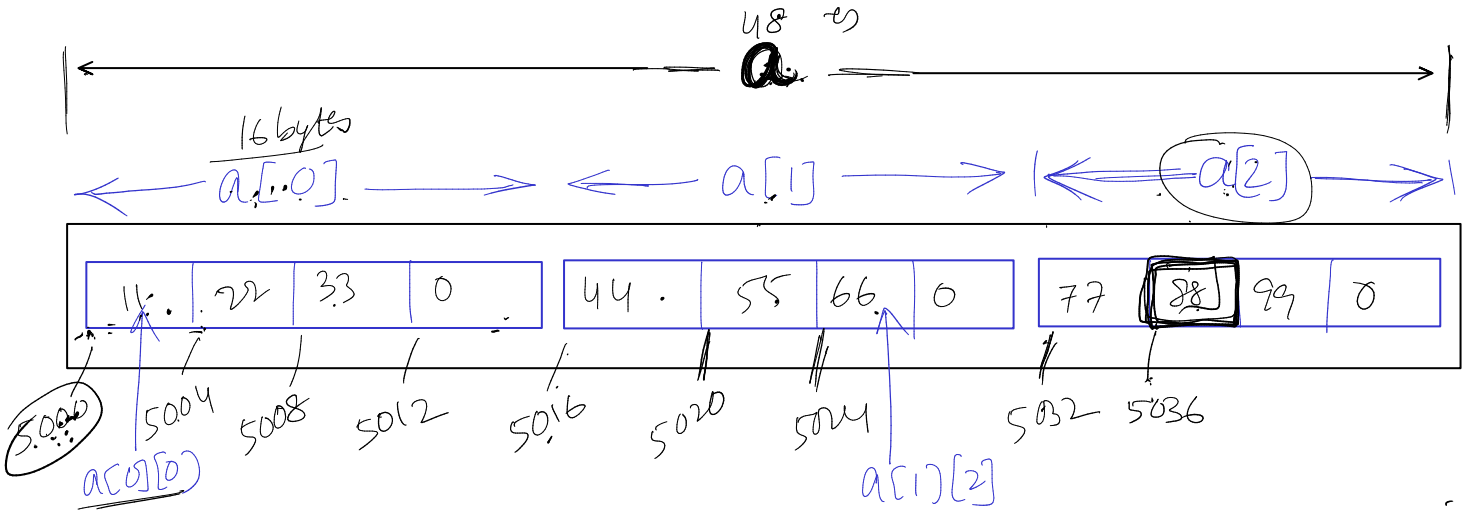
char arr[10];
double arr[10];
arr => 1000
arr+1 => 1008

arr[1] ⇔ *(arr+1)

*a[1]

*(a[1]+1) → a[1][1]

char *p = "Hello";
pf("%s", p); // Hello
"%s", *&p
p 5000
6600



int **p = a;

int *q = a[0];

int r = a[0][0];

int **p

5000

pf("%d", p[0][0]);

seg fault

---

char str[20];

float arr[10];

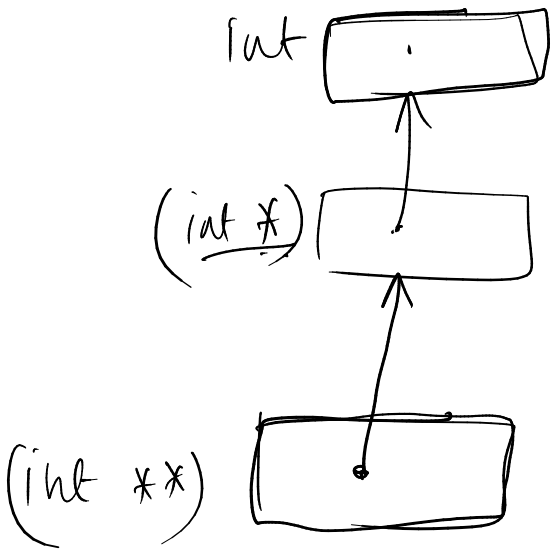
float *p = arr;

char *q = str;

Pointer variable = Array;

p = a;

A pointer to array



Pointer to Array Vs Array of Pointers

char (*p)[20];

p is pointer to Array

sizeof (p) => 8 bytes | Pointer

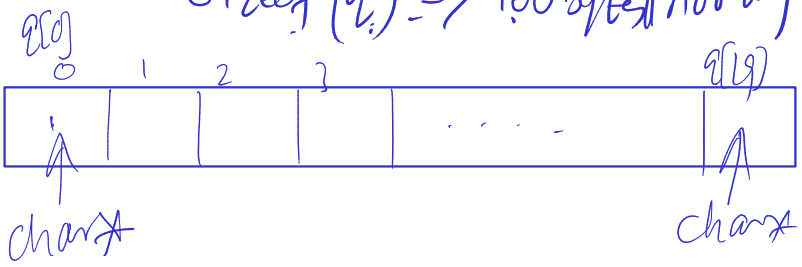
(*p) => 20 bytes | array

(**p) => 1 byte | char

char *q[20];

q is An Array of 20 pointers

sizeof (q) => 160 bytes | Array



sizeof (*q) => 8 | Pointer

sizeof (**q) => 1 | char